A certificate-based audit of simple connectivity for an explicit model associated with Wuebben's proposed exotic $S^2 \times S^2$ construction

John Clyde
VentiMath
verification@ventimath.org

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Abstract

We define an explicit marked surface-bundle surgery manifold V_{aud} modeled on the data in Wuebben's proposed exotic $S^2 \times S^2$ construction. Four hash-identified finite presentations are proved trivial by replayable derivation certificates under a published certificate specification. Automated acceptance of the stored derivations additionally assumes that at least one checker conforms to that specification; the two shipped implementations were developed within the project and have not received an independent human line-by-line audit. Separate combinatorial and geometric arguments identify the $(+,+)$ presentation with $\pi_1(V_{\text{aud}})$, including the marked clutching, normal frontiers, commonly based peripheral curves, and drilled transport relation. Consequently V_{aud} is simply connected. A logically separate theorem identifies the product-framing curves with the Lagrangian-framing classes needed for comparison with the symplectic source construction. Comparison with Wuebben's fixed surgery member is isolated as fourteen explicit Source Comparison Hypotheses, D1–D14. Each is tied to a target location, supporting evidence, and its remaining non-mechanical content. The three downstream manifold conclusions are then conditional on those hypotheses and the cited classification, symplectic, and Floer-theoretic results. A reported contrary fundamental-group computation remains unreconciled; the paper lists all live failure locations rather than assigning the discrepancy in advance. Thus the source-independent audit theorem, not the conditional transfer to the proposed exotic manifolds, is the paper's primary result.

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1 Introduction

The disputed step in Wuebben's proposed construction [1] is the fundamental group of a four-manifold piece V obtained by two Luttinger surgeries in a genus-two surface bundle. Simple connectivity of V is the input that drives the proposed exotic $S^2 \times S^2$, quotient, and $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$ conclusions. A contrary computation attributed to Lidman and Piccirillo has been reported, but no presentation or nontriviality witness for it is publicly available in the materials examined here. The immediate mathematical question is therefore not whether a computer algebra session terminates, but whether an explicit, correctly based presentation of the intended filling is trivial.

This paper defines a marked surface-bundle surgery manifold V_{aud} independently of that dispute and proves

$$\pi_1(V_{\mathrm{aud}}) = 1.$$

The proof has two parts. First, finite derivation certificates prove that four hash-identified presentations are trivial. Second, a geometric argument reconstructs the marked bundle, torus exteriors, peripheral curves, and drilled transport relation and identifies the fixed $(+,+)$ presentation with $\pi_1(V_{\mathrm{aud}})$. The other three sign sheets are algebraic robustness checks, not three additional geometric manifolds.

The source comparison is deliberately separate. Fourteen Source Comparison Hypotheses D1–D14 state exactly what must hold for the audit object to be Wuebben's fixed member. A separate framing theorem identifies the audit's product push-offs with Lagrangian-framing classes; it is needed only for that conditional smooth comparison, not for simple connectivity of V_{aud}. The exotic-manifold conclusions are consequently conditional implications rather than coequal claims of this note.

The audit object is useful even if a source-comparison hypothesis fails. It is a canonical formal reconstruction of the displayed bundle-and-surgery data, a new explicit simply connected surgery model, and a demanding test case for proof-producing topology. More broadly, the certificate format and the separation between finite derivation, geometric conversion, source comparison, and downstream citation are reusable in computations where a correct group calculation can still describe the wrong geometric object.

Validation status. The frozen Tietze transport and all theorem-critical derivation certificates replay in separate Python and Ruby implementations. Automated acceptance assumes that at least one checker conforms to the published mathematical specification; neither implementation has received an independent human line-by-line audit. The PL and symplectic arguments likewise have not received an external specialist audit. Transfer to Wuebben's named object depends on D1–D14, and the reported contrary computation remains unreconciled. Section 9 gives the full review boundary.

The paper has five evidence levels, displayed in Table 1. The first two are contributions of this note; the framing bridge is a third, logically separate theorem; the fourth is a conditional transfer to the target version; the fifth records consequences originating in Wuebben's work.

Statement	Status	Trust boundary
Finite Certificate Theorem	source-independent theorem about four byte-bound presentations	certificate specification; conformance of at least one checker for automated replay
Audit-Manifold Theorem	$\pi_1(V_{\mathrm{aud}}) = 1$	displayed geometric arguments, finite geometric artifacts, and named standard topology inputs
Framing Bridge Theorem	product-framing curves represent the corresponding Lagrangian-framing classes	relative Moser calculation, Lagrangian-neighborhood theorem, and compact push-off isotopy
Conditional Transfer Theorem	conditional identification with Wuebben's fixed member	Source Comparison Hypotheses D1–D14
Conditional target application	conditional recovery of Wuebben's three conclusions	Source Comparison Hypotheses D1–D14 and the cited classification, symplectic, and Floer-theoretic results

Table 1: The five claim levels. No implication changes one evidence type into another.

2 The audit manifold and principal statements

Write $\Sigma_{1,1}$ for an oriented once-punctured torus with based oriented spine loops α, β.

Definition 2.1 (the audit manifold)**.** Let F be a closed oriented genus-two surface with an ordered five-chain (a, b, c, d, e) whose regular-neighborhood complement is the disjoint union of two disks containing marked points p, O. Let φ_0 be the orientation-preserving involution reversing the chain, fixing p, O, and acting freely on c by a half-rotation. Let

$$\psi_0 = T_a \circ T_b,$$

with T_b applied first and both twists supported away from a collar of e. Form the marked bundle R_{aud} over $\Sigma_{1,1}$ by clutching the α- and β-mapping cylinders of φ_0 and ψ_0 to one common marked fiber.

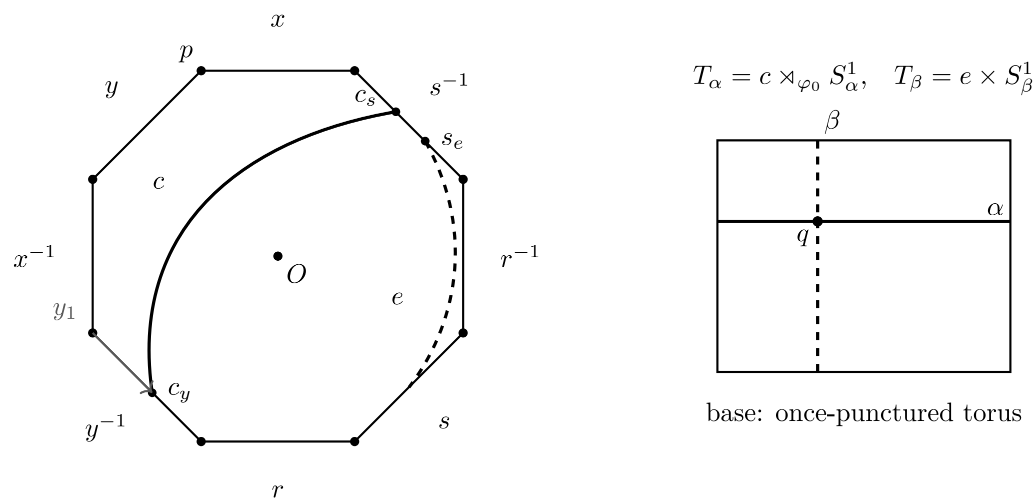
Inside R_{aud} put

$$T_\alpha = c \rtimes_{\varphi_0} S^1_\alpha, \qquad T_\beta = e \times S^1_\beta.$$

Fix the basepoint $q = (p, *)$. Let $A = [\bar{\alpha}]^{-1}$ and $B = [\bar{\beta}]^{-1}$ be the based inverse base loops. Base the oriented meridians M, N along the initial fiber arcs y_1 to c_y and s_2 to s_e, respectively. In the product normal boundaries let λ_α be the y_1-whiskered push-off that lifts the inverse α direction and closes along the selected half of c, and let λ_β be the s_2-whiskered push-off of the inverse β direction along e. Their based words Ax and $(r^{-1}Mr)B$ are conclusions of Lemmas 5.4 and 5.5, not part of this definition. Define V_{aud} by the product-framed fillings

$$M\lambda_\alpha = 1, \qquad N\lambda_\beta = 1.$$

The orientations of $M, N, \lambda_\alpha, \lambda_\beta$ are part of this definition. Thus this is the $(\epsilon_A, \epsilon_B) = (+1, +1)$ sheet in the fixed audit basis; changing one exponent while holding that basis fixed is not declared to preserve the filling.



(a) the marked fiber F (b) the two surgery tori

Figure 1: The audit manifold's marked data. Panel (a) is the symmetric octagon model of the genus-two fiber, with boundary word $xyx^{-1}y^{-1}rsr^{-1}s^{-1}$; all eight corners represent p, and O is the center. The marked curves c, e and the initial y_1 whisker determine the curve subbundles and peripheral basing. Panel (b) shows the two tori over the base loops. The drawing is explanatory; the proof uses the certified simplicial paths and incidences.

Theorem 2.2 (Audit-Manifold Theorem)**.** The manifold V_{aud} is simply connected.

Theorem 2.2 is source-independent: it concerns the explicit object in Definition 2.1. Its proof combines the finite certificate theorem below with the geometric identification in Section 5. Only the later Conditional Transfer Theorem transfers it to the object named in Wuebben's target version.

3 The finite certificate theorem

Theorem 3.1 (finite-certificate statement)**.** Let `S` be the byte string

`verification/luttinger/sealed_transport/r_presentations.json`,

whose SHA-256 digest is

`1e5fc8cb6872a25fa45a227c1e1c207b288c861a3998fe3aae8722ea510c852c`.

Read a positive integer j in `S` as the generator g_j and a negative integer $-j$ as g_j^{-1}. The top-level fields `ngens=3` and `relators` define a 3-generator, 78-relator presentation Q. Take the four entries of the top-level array `paper_fillings` whose `half_drift` field is `"n0_y1"`; these are entries 0–3, and their (`sign_a`, `sign_b`) fields run over $(+1, +1)$, $(+1, -1)$, $(-1, +1)$, $(-1, -1)$ in that order. Append the two word lists of such an entry, of lengths 94 and 132, to the 78 relators of Q, and call the resulting 3-generator, 80-relator presentation $P_{\epsilon_A, \epsilon_B}$ for $\epsilon_A =$ `sign_a` and $\epsilon_B =$ `sign_b`. Every one of these four explicitly defined presentations presents the trivial group.

Definition 3.2 (letters, words, and equation keys)**.** Fix generators $g_1, \dots, g_n$. Source relators are signed words in $\{\pm 1, \dots, \pm n\}$, with j denoting g_j. Certificate records use the monoid alphabet $L_n = \{1, \dots, 2n\}$ and the decoding map

$$\operatorname{dec}(2j-1) = j, \qquad \operatorname{dec}(2j) = -j.$$

Thus the required inverse-letter table is $(2, 1, 4, 3, \dots, 2n, 2n-1)$. If w is a signed word, write $\bar{w}$ for its reverse with all signs changed, $\operatorname{fr}(w)$ for free reduction, and $\operatorname{cr}(w)$ for the word obtained by additionally removing inverse first/last pairs until none remain. Define

$$\operatorname{ck}(w) = \min_{\text{lex}}\{\text{cyclic rotations of } \operatorname{cr}(w) \text{ and } \overline{\operatorname{cr}(w)}\},$$

with $\operatorname{ck}(1) = 1$. For certificate words u, v put

$$E(u, v) = \operatorname{ck}\big(\operatorname{dec}(u)\overline{\operatorname{dec}(v)}\big).$$

Equality of equation keys therefore means equality after free and cyclic reduction, cyclic conjugacy, and inversion. Each operation preserves the normal closure of the equation word.

Definition 3.3 (literal rewrite trace)**.** Suppose records $R_0, \dots, R_{i-1}$ have been accepted, with $R_r = (\ell_r, \rho_r)$. A trace on a word w is a list of pairs (r, p) with $r < i$ and $0 \le p \le |w|$. At that step the contiguous subword beginning at p must equal ℓ_r literally, and it is replaced by ρ_r. No cyclic matching, inversion, or implicit free reduction occurs inside a trace. Write $T(w)$ for its final word.

Definition 3.4 (certificate grammar)**.** A `luttinger-kbmag-proof-v1` certificate is a finite ordered list of records (ℓ_i, ρ_i), together with source-binding metadata and identity roots. Every side is a word over L_n. The case index must be an integer in the range of the source filling array. The source digest, case index and slug, generator count, full relator list, and inverse table must agree exactly with the hash-bound input. A record has exactly one of four proof forms.

(i) An *inverse axiom* has $\rho_i = 1$ and $\ell_i = a\,\iota(a)$ for the required inverse table ι.

(ii) An *input-relator record* names an in-range source relator q_k and satisfies $E(\ell_i, \rho_i) = \operatorname{ck}(q_k)$.

(iii) An *overlap record* names parents $a, b < i$, an integer offset d with $-|\ell_b| < d < |\ell_a|$, and two traces. The two parent left sides must agree literally on the resulting nonempty overlap. Let w be the shortest word containing both occurrences. Replacing the occurrence of ℓ_a by ρ_a gives one branch w_a, and replacing ℓ_b by ρ_b gives the other w_b. If the recorded traces give $u = T_a(w_a)$ and $v = T_b(w_b)$, validity requires $E(\ell_i, \rho_i) = E(u, v)$.

(iv) A *change record* names an old record $o < i$ and traces on its two sides. Their literal outputs must equal the separately stored `reduced_left` and `reduced_right` words, say u, v, and validity requires $E(\ell_i, \rho_i) = E(u, v)$.

Finally the root array has length exactly $2n$; its entry for each $a \in L_n$ is an in-range record whose two sides are literally $(a, 1)$. Full-inventory mode also requires one correctly slugged certificate for each of the eight source fillings, with no repeated index or slug and with the stated presentation dimensions.

Theorem 3.5 (certificate soundness)**.** Let G be the group defined by the source relators for a selected filling. If a certificate satisfies Definitions 3.2–3.4, then G is trivial.

Proof. Induct over the records. An inverse axiom is an equality in the free group. For an input record, the equation word is a freely and cyclically reduced cyclic conjugate or inverse of a defining relator, so its two sides agree in G. Every trace replaces a subword by the other side of an earlier valid equation and therefore preserves the represented element of G.

For an overlap, both untraced branches come from one source word by applying one earlier equality, so they agree in G; their traces preserve that equality. Equality of equation keys transfers the resulting equality to (ℓ_i, ρ_i). For a change, the two sides of an earlier valid equation are separately rewritten by earlier equations, and the same equation-key argument applies. Thus every accepted record is an equality in G. The root records say literally that every generator letter and its inverse equal 1, so $G = 1$. □

Definitions 3.2–3.4 are the normative mathematical specification. The complete reference control flow and concrete file format are reproduced in the supplement and in `verification/luttinger/CERTIFICATE_SPEC.md`. The Python and Ruby checkers separately implement it using only their standard libraries. They share neither source code nor runtime, but both were produced within this project and have not received an independent human line-by-line audit.

Proof of Theorem 3.1. For each of the four selected entries, the certificate bundle prepared with this version contains a certificate bound to the displayed source digest, case index, slug, complete relator list, generator count, and inverse table. The Python and Ruby implementations both accept every record and all six identity roots. The four certificates retain respectively 6672, 2470, 3803, and 3620 records in the source order $(+,+), (+,-), (-,+), (-,-)$. Hence Theorem 3.5 applies to each selected filling. Separately, two standalone checkers replay the frozen 99,860 elementary Tietze eliminations that produced `S` from its serialized raw input. No geometric interpretation of either byte string enters this proof. □

The finite theorem is therefore relative only to its published mathematical specification and to the proposition that at least one implementation conforms to it. The rest of the paper asks the different question: what geometric object do the hash-bound presentations describe? The fixed $(+,+)$ presentation is identified with V_{aud} below; comparison with Wuebben is postponed to Section 7. The supplement records the complete implementation inventory and the detailed layer-by-layer trust boundary.

4 Geometric identification of the audit manifold

This section proves that the $(+,+)$ finite presentation of Theorem 3.1 is a presentation of the explicitly defined V_{aud}. The other three sign sheets remain algebraic robustness checks. This section makes no comparison with the target paper's denotation; that logically separate step is Section 7.

The genus-2 fiber is realized as a regular neighborhood of the specified five-chain of curves — five plumbed annular bands and two cone discs — with the hyperelliptic-type involution φ_0 realized simplicially; its fixed points, the curve actions $a \leftrightarrow e$, $b \leftrightarrow d$, and the free half-rotation of the middle curve c are executable assertions about the complex, not assumptions. The bundle R_{aud} is assembled from simplicial mapping cylinders over the once-punctured torus base; the two surgery tori T_α, T_β arise as induced subcomplexes, and the assembled 4-complex passes manifold checks down to the vertex-link level.

Theorem 4.1 (geometric artifact theorem)**.** For each row of Table 2, the checkers listed in the companion supplement accept the corresponding frozen artifacts. Acceptance proves the finite assertion in the second column under the displayed checker semantics. It does not, by itself, prove the topological or smooth conversion in the last column; those conversions are supplied in the proofs below.

Proof. Each checker parses its input, verifies the recorded source digest and dimensions, exhausts the finite incidences or identities specified in its source, and exits successfully only after the asserted predicates have all been tested. The public proof ledger binds the artifacts and checkers by digest and records an acyclic dependency from these finite facts to Theorem 5.7. Table 2 states the meaning assigned to each successful replay; the companion supplement's geometric inventory gives the exact filenames, checker sizes, and audit status. The later lemmas supply the mathematical conversion from the finite meaning to the geometric conclusion. □

Layer	Finite replay	Mathematical input still used
marked bundle	fiber, involution, clutching, tori, collars, and normal frontiers pass the recorded tests	ribbon and surface classification; relative gluing; PL local flatness
complement and periphery	cellular presentation, Tietze transport, 89 named words, common whiskers, normal circles, and drilled homotopy replay	cellular van Kampen and the geometric interpretation of the named paths
framing bridge	Moser identities, seam algebra, Weinstein coordinates, and first jets replay	Lagrangian-neighborhood theorem and compact push-off isotopy
source transfer	every mechanically comparable convention agrees	Source Comparison Hypotheses D1–D14, including the diagrammatic and smooth identifications

Table 2: Concise trusted-computing boundary. The full layer-by-layer table, with filenames and checker status, is in the supplement.

The dictionary tested before Proposition 7.4 is displayed in Table 3; it is not reconstructed from the fact that the filled groups happen to collapse. Here "literal" means a stored edge path in the triangulation, with its basepoint and orientation retained.

Paper datum	Mathematical role	Certified realization
F, p, O	genus-two fiber with two fixed marks	86-vertex closed surface; both fixed vertices pass surface- and link-level tests
a, b, c, d, e	ordered filling five-chain	independently reconstructed ribbon; oriented c at V_2 represents $(rx)^{-1}$
φ_0	$x \leftrightarrow r$, $y \leftrightarrow s$	simplicial half-turn fixing p, O, checked on based paths
$\psi_0 = T_a T_b$	$x \mapsto y^{-1}$, $y \mapsto yx$, $r \mapsto r$, $s \mapsto s$	ordered flip stack with T_b first; based monodromy replay
T_α	$c \rtimes_{\varphi_0} S^1_\alpha$	induced torus on the c-stack; local-flatness and subbundle checks
T_β	$e \times S^1_\beta$	induced product torus on the e-stack; local-flatness and product checks
A, B	$[\bar{\alpha}]^{-1}, [\bar{\beta}]^{-1}$	literal based edge paths from q, retaining their orientations
M, N	meridians based along y_1, s_2	oriented dual normal circles with the same literal whiskers; all 592 normal fibers checked

Table 3: Marked-bundle dictionary for the fixed $n = 0$ member. Peripheral longitudes, which carry the sign-sensitive part of the identification, are displayed separately in Table 4.

Lemma 4.2 (equivariant marked fiber)**.** There is an orientation-preserving homeomorphism $h : |F_K| \to F$ carrying the ordered curves (a, b, c, d, e) and marked points (p, O) to those of Definition 2.1 and satisfying $h\varphi_K = \varphi_0 h$. It carries the named based loops x, y, r, s with their displayed orientations.

Proof. Finite evidence. The independent ribbon reconstruction has

$$i(a, b) = i(b, c) = i(c, d) = i(d, e) = 1, \qquad i(u, v) = 0 \quad \text{otherwise.}$$

Its regular neighborhood retracts to a graph with six vertices and ten edges, hence has Euler characteristic -4. The two disjoint symplectic pairs (a, b) and (d, e) force genus two, so the neighborhood has two boundary components. The complement in the certified genus-two surface has Euler characteristic two and exactly those two boundary circles; it is therefore two disks. An exhaustive rotation-system check begins with the 36 cyclic orders at the independent four-valent crossings. Alternation, orientation preservation, two boundary faces, and preservation of the two faces leave exactly four systems; reversing the permitted curve orientations normalizes all four to one equivariant oriented ribbon type. The primary and independent fibers realize that same labeled type, including the involution on every directed curve end and the two p/O faces.

Topological input. Map vertex disks and edge bands by the checked cyclic correspondence. These product maps agree on attaching intervals, so they give an equivariant homeomorphism of the ribbon neighborhoods. Each complementary disk is invariant: if the involution exchanged them it would have no fixed point there, contrary to the certified two-point fixed set. The classical periodic-disk theorem identifies each nontrivial orientation-preserving order-two disk action with a half-turn. Extending over the quotient disks and lifting gives the required equivariant marked map. The based path comparison then reads, literally,

$$\varphi_0 : (x, y, r, s) \longmapsto (r, s, x, y).$$

No fundamental-group filling or downstream theorem enters this argument. □

Lemma 4.3 (relative monodromy and graph clutching)**.** The map h can be chosen so that it conjugates the two monodromy representatives relative to the full c and e collars. It consequently extends to an orientation-preserving fiberwise homeomorphism

$$H : |K| \longrightarrow (R_{\mathrm{aud}})_{\mathrm{top}}$$

carrying the basepoint, both torus subbundles, and their product collars to the corresponding marked objects.

Proof. Finite evidence. The literal based actions are

$$\varphi_0 : (x, y, r, s) \mapsto (r, s, x, y), \qquad \psi_0 : (x, y, r, s) \mapsto (y^{-1}, yx, r, s).$$

The second action is obtained from complete whiskered paths, not free homotopy classes, and is replayed in two differently subdivided beta stacks. On the three-row c collar the alpha map is exactly the free shift by four columns. The beta representative is the ordered supported word $T_a \circ T_b$, with T_b first; all 1,536 trace cells avoid the full e collar, and the 3,072 restricted tetrahedra are exactly its product staircase.

Topological conversion. Both bundle models are the union of a common fiber block and two mapping cylinders over the rank-two spine of $\Sigma_{1,1}$; there is no base two-cell. For a monodromy f, write

$$M_f = (F \times [0, 1])/((z, 1) \sim (f(z), 0)).$$

If $hf_K = f_*h$, the formula $[z, t] \mapsto [h(z), t]$ respects the seam and has the analogous inverse. For an isotopic target representative, the standard interpolating formula $[z, t] \mapsto [k_t(z), t]$, with $f_*k_1 = k_0 f_K$, gives the same conclusion. The alpha conjugacy is exact on the c collar. For beta, both sides use the identical supported twist word; interpolating twist profiles is relative to the disjoint e collar. The two handle maps restrict to h on their common fiber and therefore glue, as do their inverses. This constructs H directly; bundle classification and Dehn–Nielsen–Baer are independent controls, not proof inputs. □

Corollary 4.4 (the marked torus pair)**.** Under H, the induced subcomplexes $c \rtimes_{\varphi_K} S^1_\alpha$ and $e \times S^1_\beta$ map to T_α, T_β, together with their product collars and the p-section. All smooth and symplectic arguments may therefore be performed in the audit-defined smooth target R_{aud}; no smoothing of $|K|$ is chosen.

The model results through Corollary 4.4 concern V_{aud} and use no source-comparison clause. Five structurally different constructions test subdivision, monodromy order, twist sign, labels, and orientations. These internal controls support the audit object; they do not identify it with the object denoted by a source diagram. That comparison is made only in Section 7.

5 Peripheral data and the product filling

Let $T = T_\alpha \sqcup T_\beta$. Transport the explicit normal block neighborhood $N_{1/2}(T)$ below through the marked homeomorphism H, and use the resulting product-collared neighborhood as $\nu(T)$. Put $C_{\mathrm{aud}} = R_{\mathrm{aud}} \smallsetminus \operatorname{int} \nu(T)$. This section identifies the normal boundary of each component, its commonly based peripheral pair, and the two filling slopes. The statements are deliberately separated so that a failure of one cannot be hidden by triviality of the final groups. This section proves the intrinsic product-filling theorem; it uses no symplectic or Lagrangian-framing input.

5.1 Normal frontier and compact exterior

Lemma 5.1 (local flatness and the normal frontier)**.** The two torus subcomplexes are disjoint, full, and locally flat. The barycentric level set

$$N_{1/2}(T) = \left\{ x : \sum_{v \in T} \lambda_v(x) \geq \tfrac{1}{2} \right\}$$

is an explicit normal D^2-block bundle, and its frontier $F_{1/2}(T)$ is the normal-circle boundary. The derived frontier used by the computation is PL homeomorphic to $F_{1/2}(T)$, carrying every extracted dual loop to a literal oriented normal-circle fiber.

Proof. For every simplex $\sigma \subset T$, the complete link pair is checked to be

$$(S^3, \text{unknotted } S^1), \quad (S^2, S^0), \quad (S^1, \varnothing)$$

in dimensions $0, 1, 2$, respectively. The enumeration covers all 296 vertices, 888 edges, and 592 triangles. Vertex-link 3-spheres and unknotted link circles carry independently replayed collapse and cyclic-knot witnesses. Thus the codimension-two local pairs are standard.

In a simplex with torus face A and opposite face B, barycentric coordinates give the unique block form

$$N_{1/2}(T) \cap \sigma \cong \Delta_A \times C(\Delta_B), \qquad F_{1/2}(T) \cap \sigma \cong \Delta_A \times \Delta_B.$$

The formulas agree on common faces. If $b(s)$ is a mixed-simplex vertex of the computed derived frontier, the map

$$b(s) \longmapsto \tfrac{1}{2} b(s \cap T) + \tfrac{1}{2} b(s \setminus T)$$

extends cellwise to the canonical subdivision map for $\Delta_A \times \Delta_B$ and hence gives the global PL homeomorphism. This is checked on 113,336 mixed vertices and 769,256 derived comparabilities. For every torus triangle the alternating dual loop has constant torus coordinate and is precisely the subdivided normal link circle. No regular-neighborhood uniqueness or smoothing theorem is used. $\square$

Lemma 5.2 (complement presentation)**.** Let $L = K[V(K) \setminus V(T)]$ be the full subcomplex on the vertices outside T. There is a homotopy-equivalence diagram

$$|L| \hookrightarrow |K| \setminus \operatorname{int} N_{1/2}(T) \hookrightarrow |K| \setminus |T|, \qquad |K| \setminus \operatorname{int} N_{1/2}(T) \cong C_{\mathrm{aud}},$$

in which both spaces on the right strongly deformation retract onto $|L|$. Consequently the maximal-tree edge generators of L and its oriented triangle boundaries give a presentation of $\pi_1(C_{\mathrm{aud}})$. The raw presentation has 99,863 generators and 321,702 relators; the sealed Tietze certificate reduces it to the 3-generator, 78-relator complement presentation in the source of Theorem 3.1, while transporting all 89 named peripheral and comparison words.

Proof. On each simplex not contained in the full subcomplex T, let $r(x)$ be the point obtained by setting the coordinates on T to zero and renormalizing the remaining barycentric coordinates. The straight-line homotopy $H_t(x) = (1-t)x + tr(x)$ agrees on faces and fixes $|L|$. If $\rho(x) = \sum_{v \in T} \lambda_v(x)$, then $\rho(H_t(x)) = (1-t)\rho(x)$. Thus H_t stays in $|K| \setminus |T|$ and also restricts to the block exterior $\{\rho \leq 1/2\} = |K| \setminus \operatorname{int} N_{1/2}(T)$. The homeomorphism H and the chosen product collar identify that block exterior with C_{aud}. This proves all three asserted homotopy identifications, including the previously implicit passage from the deleted subcomplex to the compact torus exterior.

A maximal tree on L's 8,860 vertices has 8,859 edges; subtracting from the 108,722 complement edges gives the stated 99,863 generators. Collapsing the tree and applying cellular van Kampen to every oriented two-simplex gives the raw presentation. Each of the 99,860 recorded Tietze eliminations is replayed against the unique current relator that implies it; separate Python and Ruby checker implementations agree on the final presentation and every tracked word. $\square$

Based meridians and product-framing longitudes for both tori are traced as literal simplicial loops carrying the paper's own basing whiskers. The paper's sign-correction package (the corrected relations of its §8.3 and the push-off basing correction of its §8.5) is reproduced independently, and a combinatorial sweep of the relevant transport square resolves the basing double-count question in the direction the paper's corrected convention requires. Both surgery tori are certified locally flat at every simplex link, and every extracted dual meridian is checked to be a literal normal-circle fiber of an explicit normal block model, with no regular-neighborhood theorem invoked. The closed section $\widehat{\Gamma}$ of the doubled bundle — the union of the two copies of the section through a fixed point of φ_0 — is extracted as a closed orientable genus-2 surface, and its self-intersection $\widehat{\Gamma} \cdot \widehat{\Gamma} = 0$ is certified with an explicit normal push-off; this number enters the conditional consequences and the complete ledger in the supplement.

5.2 Relation sheet and product fillings

To make Theorem 5.7 inspectable without opening the repository, fix the orientation sheet used by the triangulation and put $\delta = r^{-1}$. Before filling, the paper-coordinate presentation read from the literal paths has the surface relation, the following eight transport relations, and the drilled-fiber relation:

$$\begin{aligned}
&AxA^{-1} = r, \qquad AyA^{-1} = s, \qquad && ArA^{-1} = x, \quad AsA^{-1} = Ny,\\
&BxB^{-1} = y^{-1}, \quad ByB^{-1} = M^{-1}yx, && BrB^{-1} = r, \quad BsB^{-1} = (r^{-1}M^{-1}r)s,\\
&[x,y][r,s] = 1, \qquad B(s^{-1}r^{-1}yx)B^{-1} && = r^{-1}s^{-1}x.
\end{aligned}$$

The two commonly based product-framed directions are

$$\lambda_\alpha = Ax, \qquad \lambda_\beta = (r^{-1}Mr)B,$$

and the four algebraic sign sheets append exactly

$$M\lambda_\alpha^{\epsilon_A} = 1, \qquad N\lambda_\beta^{\epsilon_B} = 1, \qquad (\epsilon_A, \epsilon_B) \in \{\pm 1\}^2. \tag{1}$$

These are four algebraic specializations in one fixed symbolic basis. Only the $(+,+)$ specialization is used below as a presentation of V_{aud}. The finite presentations of Theorem 3.1 are the frozen Tietze images of all four literal word lists; for the other three, no identification with a filled geometric manifold is asserted here.

Lemma 5.3 (peripheral convention changes)**.** Let (μ, λ, ν) be an oriented integral basis of the boundary three-torus of one deleted torus, with geometric filling slope $s = \mu + \lambda$. Reversing the orientations used to name both μ and λ, while leaving ν fixed, is the coordinate change

$$D_- = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in SL(3, \mathbb{Z}).$$

It carries s to $-s$ and hence preserves the unoriented filling slope. The corresponding boundary map extends across the filling $T^2 \times D^2$. In the commonly whiskered peripheral subgroup, the same relation is written

$$(\mu\lambda)^{-1} = \mu^{-1}\lambda^{-1} = 1.$$

By contrast, reversing only the displayed longitude changes the slope to $\mu - \lambda$; the matrix $\mathrm{diag}(1, -1, -1) \in SL(3, \mathbb{Z})$ sends s to that distinct primitive class. It is not a convention change preserving the fixed filling unless an additional self-diffeomorphism of the complement carrying $\mu + \lambda$ to $\mu - \lambda$ is supplied.

Proof. The two displayed matrices give the stated changes on $H_1(T^3; \mathbb{Z})$. A filling is determined by the unoriented primitive class that bounds the meridian disk, so s and $-s$ give the same filling. A boundary automorphism preserving that class up to sign descends on the quotient by $\langle s \rangle$ and extends over $T^2 \times D^2$ by the corresponding torus automorphism and, when necessary, reversal of the meridian disk. Because the meridian and longitude use a common whisker, their boundary subgroup is abelian; hence inversion of $\mu\lambda$ gives the displayed word. Finally $(1, -1, 0)$ is not $\pm(1, 1, 0)$, proving the last assertion. The same argument applies independently on the alpha and beta boundary components. □

Direction	Paper word	Stored based loop	Independent evidence
λ_α	Ax	`lb_a_y1`, based by y_1 at c_y	literal path construction; 2,506-record complement certificate
λ_β	$(r^{-1}Mr)B$	`lb_b_s2`, based by s_2 at s_e	punctured transport square and independent extraction; 1,540-record complement certificate

Table 4: The two sign-sensitive longitude identifications. Neither certificate contains a filling relation. The words are oriented in the fixed audit basis; Lemma 5.3 states the complete coordinate change required to preserve the filling.

5.3 Peripheral curves and drilled transport

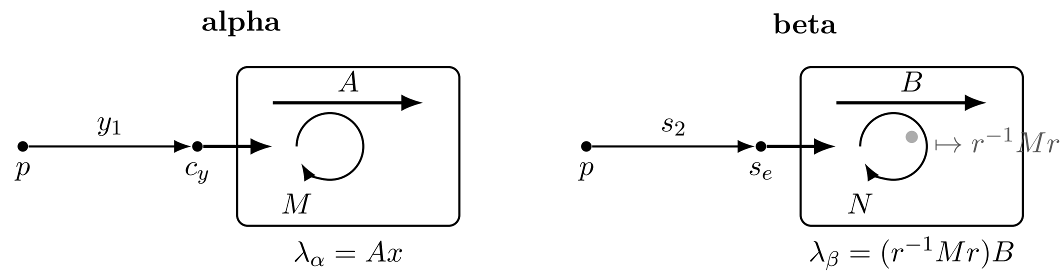


Figure 2: Common-whisker peripheral construction. On the alpha side the meridian and longitude use the same y_1 path to c_y. On the beta side they use the same s_2 path to s_e; the swept basing square meets T_α once, and puncturing that intersection contributes the conjugated meridian $r^{-1}Mr$. The diagram records basing and word order, not metric geometry.

Lemma 5.4 (the alpha peripheral pair)**.** Under the normal-frontier identification of Lemma 5.1, the stored loops `geom_M` and `lb_a_y1` are the commonly based meridian and product longitude (M, λ_α), and

$$\texttt{lb\_a\_y1} = Ax \quad \text{in } \pi_1(C_{\text{aud}}).$$

The displayed equality is certified using only the 78 complement relators, with neither filling relation present.

Proof. Consider T_α. The curve c meets the oriented y edge once, at c_y. Starting at p, the stored path follows the initial segment y_1 to c_y, makes a normal jog to the frontier of the explicit normal block, and traverses the oriented dual normal circle. Returning along the same whisker gives M. The longitude uses that identical whisker and normal jog, traverses the base direction A, and closes along the selected half of c. The stored words are `geom_M` and `lb_a_y1`; they are what M and λ_α denote throughout, and they are what Theorem 3.1 literally fills with.

That the two loops use the same whisker is not an inference but a fact one can read off the sealed transport input: there $\texttt{geom\_M} = y_1\, \mu_\alpha\, y_1^{-1}$ and $\texttt{lb\_a\_y1} = y_1\, \lambda_\alpha^\circ\, y_1^{-1}$ with the *identical* four-edge whisker $y_1 = e_{24046}e_{90328}e_{79763}e_{72645}$, seven- and six-edge circles $\mu_\alpha, \lambda_\alpha^\circ$, and fifteen and fourteen letters in all. So no independent conjugation is hidden in the filling word $M\lambda_\alpha^{\epsilon_A}$.

The identification of the stored word `lb_a_y1` with the paper's symbol Ax has two independent supports. Geometrically it follows from how the path is built — based on the y rail at the unique crossing c_y, whiskered by y_1, jogged to the normal-block frontier, lifting the base generator A, and closed along the selected half of c. Algebraically, the elementary elimination pass over the sealed presentation leaves the 72-letter residual $\texttt{lb\_a\_y1}^{-1}Ax$, but a proof-producing completion reduces that exact word to the identity in 61 steps. Independent Python and Ruby verifiers replay the complete 2,506-record ancestry cone from the sealed three-generator, 78-relator complement; neither filling relation is present. The frozen source, certificate, compiler, and both verifiers are in `verification/luttinger/alpha_residual/`. The supplement records the superseded development history; only the sealed certificate is used here. ☐

Lemma 5.5 (the beta peripheral pair)**.** Under the same normal-frontier identification, the stored loops `geom_N` and `lb_b_s2` are the commonly based meridian and product longitude (N, λ_β), and

$$\texttt{lb\_b\_s2} = (r^{-1}Mr)B \quad \text{in } \pi_1(C_{\text{aud}}).$$

Again the equality is certified in the unfilled complement.

Proof. The beta-coordinate word `lb_b_s2`$^{-1}(r^{-1}Mr)B$ likewise leaves a 113-letter residual under elementary elimination. The same proof-producing complement system reduces that exact word to the identity in 82 steps; separate Python and Ruby verifier implementations replay its 1,540-record ancestry cone. The frozen source, certificate, compiler, and both verifiers are in `verification/luttinger/beta_residual/`. Thus both displayed coordinate identities are algebraically certified in the complement, although Theorem 3.1 remains more direct: it fills with the literal stored boundary longitudes rather than relying on either coordinate substitution.

The beta extraction is the sign-sensitive case. The corresponding path uses the initial segment s_2 to the unique crossing s_e, the base loop B, and the fiber-normal push-off. Its swept basing square crosses T_α once. Puncturing that square contributes the conjugated meridian $r^{-1}Mr$, which is why the paper's word for this direction is $(r^{-1}Mr)B$ and not B. The stored words are `geom_N` and `lb_b_s2`, exported without substitution; they are what N and λ_β denote, and the identification with $(r^{-1}Mr)B$ is certified in the complement exactly as in the α case. An independently written extractor recovers both pairs without importing the complement, sweep, or peripheral modules. Local link checks certify both tori at every simplex, and the explicit block-frontier map checks every dual loop as a normal-circle fiber; these are the finite facts behind the geometric words "meridian" and "push-off." □

Lemma 5.6 (the drilled-fiber relation)**.** The based relation

$$B(s^{-1}r^{-1}yx)B^{-1} = r^{-1}s^{-1}x$$

holds in $\pi_1(C_{\mathrm{aud}})$, not merely in the unpunctured mapping cylinder.

Proof. The transported loop κ_3, of edge length 31, misses c but meets e once; sweeping it along the core of the beta curve would therefore meet T_β and would not prove the stated complement relation. The paper uses the specified parallel push-off. Along that push-off, the complete mapping-cylinder stack — 17,664 tetrahedra on 2,966 vertices — has no vertex on either surgery torus. A 2×35 simplicial homotopy grid at p joins the push-off basepoint loop to the core one and identifies it with the presentation's whiskered generator; it too is disjoint from both tori. The based monodromy target then reduces to $r^{-1}s^{-1}x$.

The finite checker exhausts the stack, both torus vertex sets, every grid cell, and the target path. Thus the entire transport homotopy lies in C_{aud}. This is the construction-specific fact that was absent from the earlier mapping-cylinder-only calculation; no group simplification or low-index fingerprint is used to supply it. □

5.4 Simple connectivity of the product-filled audit model

Theorem 5.7 (audit-model identification)**.** For the fixed audit sheet $(\epsilon_A, \epsilon_B) = (+1, +1)$, there is a natural surjection

$$P_{+,+} \twoheadrightarrow \pi_1(V_{\mathrm{aud}}).$$

In fact the map is an isomorphism. No geometric identification of $P_{+,-}, P_{-,+}, P_{-,-}$ is part of this statement. No clause of Source Comparison Hypotheses D1–D14 enters this theorem.

Proof. Lemmas 4.2 and 4.3 construct the marked bundle homeomorphism and carry the two curve subbundles and their product collars to the specified tori. Lemma 5.1 identifies their normal boundaries and meridians. Lemmas 5.4 and 5.5 identify the two commonly based product longitudes, including the conjugated meridian contributed by the punctured beta sweep. Lemma 5.6 places the final transport relation in the drilled complement.

By Definition 2.1, attaching the two product-framed filling pieces adds exactly the normal closure

$$\langle\langle M\lambda_\alpha, N\lambda_\beta\rangle\rangle.$$

Cellular van Kampen therefore gives the asserted surjection from the filled presentation. Lemma 5.2 gives an actual presentation of the complement, not merely a list of valid relations; applying van Kampen with the two filling relations proves that the surjection is an isomorphism. Changing a common whisker simultaneously conjugates both elements of a peripheral pair and does not change their normal closure. Lemma 5.3 records the different, simultaneous meridian–longitude reversal that preserves a filling slope. This completes the identification with Definition 2.1 without using a symplectic form, a Weinstein chart, or any comparison with Wuebben's text and figures. □

Proof of Theorem 2.2. Use the $(+,+)$ sheet in Theorem 5.7. Its source is trivial by Theorem 3.1; a quotient of the trivial group is trivial. Notice that this conclusion uses only the surjective part of Theorem 5.7, no symplectic or Lagrangian-framing result, and no source-identification assumption. □

6 The product-to-Lagrangian framing bridge

The preceding proof of $\pi_1(V_{\text{aud}}) = 1$ is complete before this section begins. The next theorem is not used in Theorem 5.7 or Theorem 2.2. It is needed only in Lemma 7.3, where the product-framed audit filling is compared with the Lagrangian surgery in the source construction.

Theorem 6.1 (product framing equals Lagrangian framing)**.** For the normalized Thurston symplectic form on R_{aud}, the product-framed push-offs in Lemmas 5.4 and 5.5 represent the canonical Lagrangian-framing classes of T_α and T_β. In particular neither acquires a meridian component when transported into a Weinstein neighborhood.

Proof. The proof takes place in the already smooth target R_{aud}. By Corollary 4.4, the marked homeomorphism H transports the product collars and their push-offs from the PL model into that target. No smooth structure is placed on the source complex. A framing push-off below means the isotopy class, in $\nu(T_i) \setminus T_i$, of a sufficiently small copy of a curve on T_i.

Relative normal form. On an annular fiber collar write

$$\Omega = f(\theta, t)\, dt \wedge d\theta, \qquad f > 0,$$

with core $t = 0$. Put

$$g(\theta, t) = \int_0^t (f(\theta, u) - 1)\, du, \qquad H_s(\theta, t) = t + sg(\theta, t).$$

Then $\partial_t H_s = (1-s) + sf > 0$. On one collar common to the compact core, H_s therefore has a smooth inverse in the radial coordinate. If $H_s(\theta, T_s(\theta, t_0)) = t_0$, then $T_0 = t_0$, $T_s(\theta, 0) = 0$, and

$$\partial_s T_s = -\frac{g(\theta, T_s)}{(1-s) + sf(\theta, T_s)}.$$

Thus $(\theta, t_0) \mapsto (\theta, T_s)$ is the relative Moser isotopy, obtained by monotone inversion rather than an abstract flow theorem. At $s = 1$, differentiation in t_0 gives $f(\theta, T_1)\partial_{t_0} T_1 = 1$, hence $T_1^*\Omega = dt_0 \wedge d\theta$. The exact identities, fixed-core condition, and a uniform collar bound are replayed in `moser_cumulative_flow.py`.

For the alpha collar the deck involution is $\tau(\theta, t) = (\theta + \pi, t)$ and the quotient coordinate is $\bar{\theta} = 2\theta$; write π_α for the quotient map. The preceding radial map is defined directly upstairs by replacing $\bar{\theta}$ with 2θ in its coefficient. Periodicity makes it τ-equivariant, so no covering-lift choice is involved. Moreover

$$\pi_\alpha^*(dt \wedge d\bar{\theta}) = 2\, dt \wedge d\theta = d(2t) \wedge d\theta,$$

which is the required factor-two normalization. These assertions are replayed in `equivariant_moser_lift.py`.

The two charts. Near T_β, ψ_0 is the identity near e, and the preceding normalization gives

$$\omega = dt \wedge d\theta_1 + K\, du \wedge d\theta_2, \qquad K > 0;$$

the cotangent momenta are (t, Ku). The in-fiber and base-normal push-offs are therefore constant-momentum copies. Near T_α, the mapping-torus seam is

$$(\theta_1, t, 2\pi, u) \sim (\theta_1 + \pi, t, 0, u).$$

On its quotient put

$$\Theta_1 = \theta_1 - \tfrac{1}{2}\theta_2, \quad \Theta_2 = \theta_2, \quad P_1 = t, \quad P_2 = \tfrac{1}{2}t + Ku.$$

These functions respect the seam and satisfy

$$dP_1 \wedge d\Theta_1 + dP_2 \wedge d\Theta_2 = dt \wedge d\theta_1 + K\, du \wedge d\theta_2.$$

The fiber push-off has momentum $(t_0, t_0/2)$ and the half-drifting base push-off has momentum $(0, Ku_0)$, both constant. Hence neither acquires a meridian component, which is precisely the equality of framings used in (1). The Liouville primitive $t\, d\theta_1 + Ku\, d\theta_2$ forces these momenta by pairing with the Θ-frame; `weinstein_chart_check.py` replays that derivation, the seam, exterior algebra, and a second opposite-shear chart.

T_β coordinates $(\theta_1, \theta_2; t, Ku)$ $\xrightarrow{\text{product}}$ constant momenta: fiber: $(t_0, 0)$ base-normal: $(0, Ku_0)$

T_α seam coordinates $(\theta_1, \theta_2; t, u)$ $\xrightarrow{\text{shear}}$ cotangent coordinates: $\Theta_1 = \theta_1 - \frac{1}{2}\theta_2,\ P_1 = t$; $\Theta_2 = \theta_2,\ P_2 = \frac{1}{2}t + Ku$

fiber push-off: $(P_1, P_2) = (t_0, t_0/2)$
half-drift push-off: $(P_1, P_2) = (0, Ku_0)$

Figure 3: The two local Weinstein-coordinate calculations. Product push-offs become constant-covector copies in both charts, so the path from a push-off to the torus never winds in a normal circle. The chart-independence argument below shows that this boundary framing class is intrinsic.

Here is the boundary-class assertion in coordinates. Let γ be either named curve, let η complete it to an oriented basis of the zero-section torus, and let μ be the positively oriented circle in a cotangent fiber. After radial projection to the unit-covector boundary, a constant nonzero covector p_0 gives

$$s \longmapsto \big(\gamma(s), p_0/\|p_0\|\big).$$

Its normal-circle coordinate is constant, so its class in the boundary basis (γ, η, μ) has μ-coefficient zero. This is the precise sense in which "constant covector" excludes a meridional component. The seam map and the quotient map above preserve the displayed orientations: their pullbacks give the positive forms written there, and the chosen positive normal circle is the boundary orientation of the ordered momentum plane.

Independence of the Weinstein chart. It remains to show that the constant-momentum conclusion is a framing class rather than a feature of the displayed coordinates. Given two Lagrangian-neighborhood charts, first compose with a cotangent lift so that their restrictions to the zero section agree. Their transition is then a symplectomorphism germ $\Phi : (T^*T, T) \to (T^*T, T)$ fixing the zero section pointwise. With fiber dilation $\delta_r(q, p) = (q, rp)$, define

$$\Phi_r = \delta_r^{-1}\Phi\delta_r \quad (0 < r \leq 1).$$

Although $\delta_r^*\omega = r\omega$, the two factors cancel, so every Φ_r is symplectic. Along the zero section the symplectic condition gives

$$d\Phi = \begin{pmatrix} I & A \\ 0 & I \end{pmatrix}, \qquad A = A^{\mathrm{T}}.$$

Taylor expansion shows that Φ_r extends at $r = 0$ by the identity. Hence Φ_r is an isotopy of germs, relative to T, from the identity to the chart transition. To make the punctured-neighborhood assertion uniform, choose a compact germ domain over the entire named curve and then choose one nonzero constant covector copy small enough that every Φ_r is defined on it. Each Φ_r fixes the zero section and is injective, so no point of that copy can meet the zero section. Compactness of the copy times $[0, 1]$ then gives a positive distance from the zero section throughout the isotopy. Thus the whole isotopy lies in one $\nu(T) \setminus T$ and proves that the two push-offs have the same boundary class. The first-jet symplectic calculation is replayed in `weinstein_chart_independence.py`. The potentially dangerous momentum shift by a closed one-form could alter the pushed-off boundary class, but it moves the zero section unless the form vanishes and is therefore excluded here. The standard Lagrangian-neighborhood theorem, in the form used by [4, §2.1 and Proposition 2.2], supplies the existence of the charts; the argument above supplies the precise chart-independence consequence needed here. □

7 Source-comparison hypotheses and conditional transfer

The finite certificate and Audit-Manifold Theorems are complete before this section begins, and neither uses the Framing Bridge Theorem. That theorem is invoked below only to compare the product filling with the target's Lagrangian surgery. The remaining question is not "authorial intent" but a precise comparison of two mathematical specifications. We therefore state the comparison assumptions individually and classify them by evidentiary type.

Definition 7.1 (source comparison hypotheses)**.** The Source Comparison Hypotheses are clauses D1–D14 in Table 5. Their conjunction is the assumption used by the Conditional Transfer Theorem. The table states their required mathematical content and evidentiary type. Exact target locations, mechanical support, and the residual assertion for every clause are recorded in the supplement; supporting evidence does not discharge a hypothesis.

Table 5: Compact Source Comparison Hypotheses. Every row remains an assumption of the Conditional Transfer Theorem; "agrees" reports evidence, not discharge. The full fourteen-row ledger is in the supplement.

Clause	Type	Required source-to-audit identification	Status
D1	textual	ordered five-chain labels and marked points	conditional; text extraction agrees
D2–D3	diagrammatic	involution, complementary marked disks, and hidden arc continuations	conditional; finite and mutation checks support
D4–D5	textual	twist order and signs; assignment of the two base holonomies	conditional; independent reconstructions agree
D6	diagrammatic	the two depicted curve subbundles are the audit tori, relative to their collars	conditional; subbundle checks support
D7–D8	mixed	path composition, base generators, oriented meridians, and named basing arcs	conditional; peripheral extractors agree
D9–D11	textual	reference half-drift, fixed member and slopes, and the fixed $(+, +)$ audit sheet	conditional; relation sheets agree
D12	smooth	source Lagrangian tori are smoothly identified with the audit tori	assumed by transfer
D13	smooth	source coefficients and framing classes give the audit's two attaching maps	assumed by transfer

D14	smooth	marked bundle equivalence upgrades to a relative orientation-preserving diffeomorphism	assumed by transfer

The textual rows can be checked against the pinned source as transcription claims. The diagrammatic rows remain identifications of depicted objects. Most importantly, D12–D14 are smooth and framing comparison hypotheses: their cited and mechanical support does not independently construct the relative diffeomorphism or prove that it is the one denoted by the source. Splitting the former four omnibus assumptions into these three evidence classes prevents a verified textual choice from being bundled with a genuinely conditional smooth identification.

Lemma 7.2 (extension across a torus filling)**.** Let $X = T^2 \times D^2$ and order an integral basis of its boundary as $(\mu, \gamma_1, \gamma_2)$, where μ bounds the disk fiber. A boundary diffeomorphism $f : \partial X \to \partial X'$ extends over X if and only if $f_*(\mu) = \epsilon\mu'$ for $\epsilon \in \{\pm 1\}$. In these bases its matrix then has the form

$$[f_*] = \begin{pmatrix} \epsilon & u_1 & u_2 \\ 0 & a_{11} & a_{12} \\ 0 & a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} \epsilon & u \\ 0 & A \end{pmatrix}, \qquad A = (a_{ij}) \in GL(2, \mathbb{Z}), \quad u = (u_1, u_2) \in \mathbb{Z}^2.$$

Consequently, a diffeomorphism of torus exteriors that carries each killed attaching slope to the corresponding slope up to sign extends across the filled pieces; the two unfilled boundary directions are accounted for by A and u.

Proof. Necessity follows because the kernel of $H_1(\partial X) \to H_1(X)$ is the primitive subgroup generated by μ. For sufficiency, write $x \in \mathbb{R}^2/\mathbb{Z}^2$ for the base coordinate and $z \in D^2 \subset \mathbb{C}$ for the disk coordinate. Up to translations, the displayed matrix is realized by

$$(x, z) \longmapsto \begin{cases} (Ax, e^{2\pi i \langle u, x\rangle} z), & \epsilon = +1, \\ (Ax, e^{2\pi i \langle u, x\rangle} \overline{z}), & \epsilon = -1. \end{cases}$$

This is a diffeomorphism of $T^2 \times D^2$ and restricts to the prescribed isotopy class on its boundary. A diffeomorphism of T^3 is isotopic to the affine representative with the same integral homology action; extending that isotopy through a boundary collar adjusts the displayed extension to the original f. For filled exteriors, conjugate the exterior boundary map by the two attaching maps. The hypothesis on the killed slope says exactly that the conjugated map sends the filling-piece meridian to $\pm\mu'$, so the preceding construction applies. This also exhibits the otherwise implicit third boundary direction. □

Lemma 7.3 (relative smooth source bridge)**.** Under D1–D14, the marked bundle comparison is represented by an orientation-preserving diffeomorphism relative to the boundary marking and the two torus collars. It carries the two product-framed attaching maps of Definition 2.1 to the two Lagrangian-framed attaching maps of the target fixed member, and therefore extends across the two surgeries.

This lemma packages the consequences of the smooth comparison hypotheses. In particular it does not independently prove D14, whose relative diffeomorphism is one of its assumptions.

Proof. Clause D14 supplies the relative smooth representative of the marked bundle equivalence. Clauses D6, D8, and D12 carry the torus collars, common whiskers, and oriented meridians. Clause D13, together with Theorem 6.1, identifies the two longitude classes and surgery coefficients. Thus the boundary diffeomorphisms of each removed $T^2 \times D^2$ carry meridian and attaching slope to meridian and attaching slope. Lemma 7.2, including its full boundary matrix, extends those maps over each $T^2 \times D^2$ filling piece and gives the claimed diffeomorphism of filled manifolds. □

7.1 The reported contrary computation and live failure locations

Wuebben's public repository reports that Lidman and Piccirillo obtained a fundamental-group calculation disagreeing with his, currently summarized as $\pi_1(V) \neq 1$. Their published paper [3] proves $H_1(V) = 0$ and normal generation by $\pi_1(F)$ but does not publish this contrary presentation. No relation sheet, generator dictionary, or nontriviality witness for the reported calculation is available in the materials examined here.

The finite Theorem 3.1 is unaffected because its objects are explicit strings. If the contrary calculation concerns the same geometric member, the live explanations are deliberately left neutral: a clause of Source Comparison Hypotheses D1–D14 may fail; a displayed geometric identification or geometric checker may be defective; the contrary calculation may be defective; or the two computations may concern different members or conventions. The absence of an external PL and symplectic audit prevents assigning the discrepancy to one location in advance. Relation-by-relation reconciliation requires the missing artifact.

Proposition 7.4 (Conditional transfer theorem)**.** Under Hypotheses D1–D14, the marked smooth surgery data defining V_{aud} agree with the data defining Wuebben's fixed $(m, n) = (0, 0)$ member. The resulting filled manifolds are orientation-preservingly diffeomorphic, with their boundary marking, peripheral curves, and framing classes identified.

The proposition is a transfer theorem under explicit comparison hypotheses, not an independent verification that the source and audit objects coincide.

Proof. Clauses D1–D3 and Lemma 4.2 identify the ordered marked fiber, the involution, and the complementary marked disks. Clauses D4–D6, Lemma 4.3, and Corollary 4.4 identify the two relative monodromies, the marked bundle, the two torus subbundles, and their collars. Clauses D7–D11 and Lemmas 5.1, 5.4, 5.5, and 5.6 identify the based meridians, product longitudes, drilled transport relation, fixed $(+, +)$ sheet, and fixed member. Finally D12–D14 and Lemma 7.3 promote the relative comparison to the smooth category and carry the two attaching maps across the surgeries. The resulting diffeomorphism is the one asserted. □

Corollary 7.5 (conditional simple connectivity)**.** Under Hypotheses D1–D14, Wuebben's fixed $(m, n) = (0, 0)$ member is simply connected.

Proof. Combine Theorem 2.2 with Proposition 7.4. □

8 Conditional application to the target construction

This section records why the audit matters to the proposed construction; it is not part of the source-independent Audit-Manifold Theorem. Let V, σ, f, Z, W, Z'' denote the fixed member, involution, regluing map, and three manifolds in [1]. Source Comparison Hypotheses D1–D14 are required even to identify V with V_{aud}, and the conclusions below also use named classification, symplectic, and Floer-theoretic results. Their exact statements, hypothesis checks, and the full dependency audit are collected in the companion supplement.

Corollary 8.1 (conditional application)**.** Assume Source Comparison Hypotheses D1–D14 and the classification, symplectic, and Floer-theoretic results cited in the companion supplement in the forms stated there. Then:

(A) $Z = V \cup_\sigma V$ is homeomorphic but not diffeomorphic to $S^2 \times S^2$;

(B) $W = V/\sigma$ is homeomorphic to Kawauchi's manifold B, while the figure-eight knot is smoothly slice in B and not smoothly slice in W, so (B, W) is a homeomorphic non-diffeomorphic pair;

(C) $Z'' = V \cup_{f \circ \sigma} V$ is homeomorphic but not diffeomorphic to $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$.

Proof. Under Source Comparison Hypotheses D1–D14, Proposition 7.4 and Corollary 7.5 supply the marked identification and simple connectivity required by all three arguments. The supplement gives the proposition-by-proposition deductions: Freedman and symplectic Kodaira dimension for (A), Hambleton–Kreck and the Klug–Levine obstruction for (B), and the odd-form classification plus the cited Floer-theoretic mixed invariant for (C). It states every external input and checks its hypotheses; the machine ledger verifies only the dependency bookkeeping. For clarity about the polarity in (B): The figure-eight knot is smoothly slice in B by construction. It is not smoothly slice in W by the relative Rochlin–Arf obstruction recorded in the supplement. $\square$

9 Limitations and independent review

The project has three principal limitations. First, both theorem-critical filled-group checkers were produced within the same AI-assisted development process. Separate languages and implementations, negative controls, and the published mathematical specification reduce ordinary software risk but do not constitute an independent human line-by-line audit. Second, the marked fiber, peripheral, and framing arguments have been decomposed above so that a PL or symplectic topologist can audit them locally, but no such external audit is claimed. In particular Theorem 6.1 is the highest-risk internal geometric bridge. A specialist review should check (i) that the named product push-offs land in the exact boundary classes used by the surgery convention, (ii) the seam, quotient, and normal-circle orientations, (iii) the zero meridional coefficient of the constant-covector copies, (iv) the uniform punctured-neighborhood isotopy in the chart-independence argument, and (v) that the relative smoothing hypotheses used in the source comparison are sufficient. The displayed calculations and finite checks make those questions local; they do not substitute for that external review. Third, the comparison with Wuebben's fixed member remains conditional on every clause of Source Comparison Hypotheses D1–D14. The downstream corollary additionally depends on its cited external results. The reported contrary calculation in Section 7.1 is therefore an active reason to scrutinize both the source comparison and the displayed geometric arguments; it is not evidence against a particular one until its underlying presentation and dictionary are available.

The 100-case framing scan is not an answer to either limitation. All sampled shifted presentations were reported trivial, so the scan does not discriminate the correct framing and has no logical role in Theorem 5.7 or Proposition 7.4. It is retained only as computational history in the supplement.

10 Conclusion

The revision isolates three logically ordered achievements. First, four hash-identified presentations are trivial under a complete published certificate specification, with the raw transport and all retained derivations replayable from frozen inputs. Second, the fixed $(+,+)$ presentation identifies the fundamental group of the explicitly defined V_{aud}, which is therefore simply connected. Third, a separate theorem identifies the two product-framing curves with their Lagrangian-framing classes. That third result plays no role in simple connectivity; it enters only the conditional comparison with the symplectic source construction. The marked fiber, relative clutching, torus collars, normal frontier, complement presentation, two peripheral pairs, drilled-fiber relation, and framing bridge remain separate results whose finite and conventional inputs are visible.

The remaining construction-specific uncertainty is Source Comparison Hypotheses D1–D14: whether the source text and diagrams define exactly the marked smooth data used for V_{aud}. The source comparison and the unavailable contrary computation meet at that boundary. The artifacts make its clauses finite and inspectable; they do not promote source interpretation into a certificate.

Even if a clause of Source Comparison Hypotheses D1–D14 fails, the source-independent content does not disappear. The paper still supplies a complete certificate theorem for four explicit presentations, an explicit simply connected audit manifold, a proof-producing transport from a large cellular presentation, and a reusable separation between finite computation, geometric conversion, source comparison, and downstream citation. What would fail is the transfer of those results to Wuebben's named manifold, not the results about the audit object itself.

11 Data and reproducibility

The target is Wuebben's arXiv:2608.17267v1 (18 August 2026), whose PDF has SHA-256

`16a6cef4998699f76ee508e062f8192cf80eeb478b7e077bfa35ccc25350186a`.

The immutable Lidman–Piccirillo v1 source digest and all interpretation inputs are recorded in `TARGET.md`. The theorem artifacts, frozen inputs, certificate checkers, manifests, run transcripts, and reconstruction programs are in the public repository and its immutable v2.0.1 archival snapshot [2]:

`https://github.com/VentiMath/exotic-s2xs2-verification`,

`https://doi.org/10.5281/zenodo.22171309`.

The version DOI identifies the exact v2.0.1 verification snapshot; the concept DOI `https://doi.org/10.5281/zenodo.22169753` resolves to the newest archived version.

The submission release has three intentionally different replay levels. The certificate-only path reads frozen files and requires only standard Python and Ruby runtimes. The geometric-audit path recomputes the finite marked-fiber, torus, frontier, peripheral, and framing checks. The full reconstruction path additionally uses the locked GAP 4.11.1 and KBMAG 1.5.9 environment and the proof-producing KBMAG patch. Commands, expected record counts, negative controls, and hashes are listed in the companion supplement. The deterministic source and archive digests are recorded in `ARXIV_SUBMISSION.md`.

Three source-extraction replays additionally require locally supplied copies of third-party material that the release does not redistribute: Wuebben's paper text, two files from his public code repository, and the Lidman–Piccirillo source figure. Their expected digests and exact retrieval instructions are in `verification/IMPORT.md`. The frozen extraction outputs, their comparison with the audit model, and every certificate used by the Audit-Manifold Theorem are contained in the release and replay without those third-party files.

The v2.0.1 release, PDFs, deterministic source archive, manifest, and clean-checkout replay were checked before that archival deposit was created. The present v2.1.0 revision changes the exposition and the formalized source comparison: it defines V_{aud} before the certificate machinery, replaces four omnibus assumptions by Source Comparison Hypotheses D1–D14, states the geometric trusted-computing boundary, makes the smooth relative bridge explicit, and rewrites the downstream result as a conventional conditional corollary. Before arXiv submission the v2.1.0 release must itself be archived and its exact version DOI added to the submission metadata. The finite filled-group certificates and the sealed transport are unchanged.

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